

Calibrating the alarm, not the interval: distribution-free notice guarantees for end-of-life alarms with censored calibration data

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ABSTRACT

An end-of-life alarm must not let a unit fail before the replacement the alarm scheduled. Conformal prediction gives remaining-useful-life intervals with distribution-free coverage, but an alarm acts at a data-dependent stopping time, where per-cycle coverage does not apply. We calibrate the alarm itself. For a debounced threshold trigger on any causal signal, each calibration unit has one critical level, the lowest warning level that would still have left the lead time, and the conformal quantile of these levels guarantees that a new exchangeable unit is replaced before failure with probability at least $1 - \alpha$. Treating censored units as failed just after they were last seen alive keeps the guarantee under any censoring; it has a training-conditional form; cross-fitting calibrates on all training units; n units cannot certify a failure probability below $1/(n + 1)$; and an online update bounds the long-run failure rate for any sequence of units. Under protocols hashed in advance, on PHM08, C-MAPSS FD001–FD004, N-CMAPSS and 123 lithium-ion cells, failure rates were 3.8, 9.3 and 19.4 % at nominal 5, 10 and 20 %, whereas per-cycle conformal lower bounds on the same signal failed on 18–70 % of units at nominal 10 % (6–71 % with three other learners). Censoring by an earlier alarm cost nothing; early censoring made it uninformative. With cross-fitting, certification cost about as much as the best uncertified rules on the same models. Across battery batches a calibrated level failed on 61–74 % of cells; online updating from the new batch's outcomes brought this to about 13 % without pilot failures.

1. Introduction

A prognostic model creates value through the maintenance actions it triggers. For a component that is replaced once, near the end of its life, the action is usually triggered by an *end-of-life alarm*: a policy watches the unit cycle by cycle and raises one alarm, which schedules a replacement after a lead time for parts, crew and access. The first reliability requirement on such an alarm is therefore not the accuracy of a remaining useful life (RUL) estimate at some cycle chosen by the evaluator, but the probability that the unit fails before the replacement the alarm scheduled. This paper gives that probability a distribution-free, finite-sample guarantee.

The usual ways of evaluating and calibrating prognostics do not control it. Accuracy metrics and asymmetric acceptance windows score an estimate at a time, and a window score at the alarm mixes the estimate with the alarm time [1, 2]. Decision-oriented metrics, such as the metric M of Kamariotis et al. [3], score and tune policies by long-run cost but give no guarantee for a new unit. Conformal prediction gives RUL intervals with distribution-free coverage [4–6], but the coverage holds at a cycle chosen independently of the data, whereas an alarm acts at the first cycle at which the estimate, or its lower bound, crosses a level: a stopping time at which the estimate is, by selection, at its most optimistic. Two further obstacles stand between any calibration and the field. Fleet data are censored, because most units are replaced before they fail; and populations change, from one production batch or operating campaign to the next.

This paper calibrates the alarm itself. For a threshold trigger on any causal signal, the notice left by the alarm is monotone in the warning level, so each calibration unit has a single *critical level*: the lowest warning level at which its alarm would still have left more than the lead time. The split-conformal quantile of these critical levels is a warning level with a finite-sample, distribution-free guarantee that a new exchangeable unit is replaced before it fails with probability at least $1 - \alpha$. The contributions are:

- Theory.** A closed-form critical level for debounced threshold triggers (Lemma 2); the notice guarantee with a matching upper bound (Theorem 1) and its training-conditional form, which says how far the failure probability of one calibrated alarm can stray from α (Corollary 1); a decision-aware choice of the level that keeps the guarantee (Corollary 2); a cost bound and a certification floor that state what n calibration units can certify (Proposition 1); a path-wise censoring bound under which the guarantee holds for *any* censoring mechanism, so that units replaced before failure can be used for calibration (Lemma 3, Theorem 2); a cross-fitted variant that calibrates on all training units (Theorem 3); and an online update that bounds the long-run failure rate for any sequence of units, without exchangeability (Theorem 4). A supporting result (Proposition 2) explains why window scores of threshold alarms measure timing and cannot certify notice.
- Method.** The conformal notice alarm (CNA) and its decision-aware, cross-fitted and online variants (Algorithm 1), which wrap any signal, including a

learned regressor and a physically motivated capacity trend.

3. **Evidence.** A protocol and four amendments, each hashed before its experiments; seven datasets (PHM08, C-MAPSS FD001–FD004, N-CMAPSS and 123 lithium-ion cells in three batches); comparisons with interval-calibrated alarms, window- and cost-tuned alarms, the probability-threshold policy of Kamariotis et al. (also in their own decision setting and metric), age replacement and a capacity-trend baseline; four learners and 27 alarm settings; three censoring mechanisms; and transfer to held-out battery batches and N-CMAPSS subsets.

The main findings are five. (i) Where calibration and new units were exchangeable, the guarantee held on all seven datasets: pooled failure rates were 3.8, 9.3 and 19.4% at nominal 5, 10 and 20%, whereas a per-cycle conformal lower bound on the same signal and trigger failed on 18–70% of units at nominal 10%. Both held with three further learners, including a neural network on raw sensor windows, and the guarantee held for each of 27 settings of the alarm. (ii) Censoring by an earlier alarm cost nothing: with half of the calibration units replaced before failure, the certified level was identical to the one computed from complete data. Censoring early in life kept the guarantee but made it uninformative. (iii) The certificate's cost premium is governed by the number of calibration units. With a 60/40 split it was large on small datasets; cross-fitting on all training units put the certified alarm on a par with the best uncertified rules built on the same models, never significantly dearer and sometimes cheaper. (iv) The guarantee did not transfer across battery batches (61–74% failures on two held-out batches). Ten pilot cells of the new batch run to failure restored it, and so did online calibration from the new batch's own outcomes without any pilot failure (13% failures over the whole sequence, 10% in its second half). (v) In the decision setting of Kamariotis et al. the certified alarm was statistically indistinguishable from their optimised probability-threshold policy on four of five datasets and better on one. Negative results, including three failed hypotheses, are reported with the same weight as positive ones.

Section 2 reviews related work, Section 3 states the problem, Section 4 gives the theory and Section 5 the method. Section 6 describes the design, Section 7 the results and Section 8 their implications and limits.

2. Related work

Evaluating prognostics through decisions. Remaining useful life (RUL) estimates are usually scored at times chosen by the evaluator, with accuracy metrics or asymmetric scores that penalise late predictions [1, 2, 7]. A line of work in this journal evaluates prognostics through the maintenance decisions they trigger.

de Pater et al. [8] schedule aircraft maintenance from alarms raised on RUL prognostics and protect against prognostic error with a safety factor; Kamariotis et al. [3] propose a metric M , the relative excess long-run cost of a policy over perfect prognostics, and tune probability-threshold policies with it; dynamic predictive-maintenance policies act on predicted failure probabilities [9, 10]. Age replacement is the benchmark that any condition-based policy must beat [11, 12]. None of these gives a finite-sample guarantee on the notice an alarm leaves.

Conformal prediction for RUL. Conformal prediction turns any point predictor into prediction sets with finite-sample, distribution-free coverage under exchangeability [13–15]. Javanmardi and Hüllermeier [4] conformalised gradient boosting and convolutional networks to obtain RUL intervals on C-MAPSS; Robinson [5] used asymmetric conformalised quantile regression [16] to make the lower RUL bound conservative; Wang et al. [6] combined conformal prediction with imputation of missing sensor data and noted that exchangeability is a concern for degradation data. In all of these the calibrated object is the interval at a given cycle. Candès et al. [17] give conformal lower bounds on survival times under censoring, at a fixed covariate value and under assumptions on the censoring mechanism. An alarm, however, is a stopping rule, evaluated at the cycle at which the bound first crosses a level; per-cycle coverage does not imply coverage at that time, and Section 7.1 shows that the difference is large.

Risk control, cross-fitting and shift. Conformal risk control [18] calibrates a scalar parameter so that the expected value of a monotone loss is bounded, and nested conformal prediction [19] expresses conformal procedures through nested families of sets. Our construction is an instance of the nested view, with the warning level as the nesting parameter and failure before the planned replacement as the loss. What is specific is the closed-form critical level of a debounced trigger (Lemma 2), the pathwise censoring bound that makes calibration possible with replaced units (Theorem 2), the decision-aware choice that keeps the guarantee (Corollary 2), and the certification floor (Proposition 1). Cross-fitting follows the jackknife+/CV+ construction of Barber et al. [20]. When calibration and new units are not exchangeable, weighted and non-exchangeable conformal methods bound the coverage gap [21, 22], but they need a model of the shift; online methods instead track the target error rate under arbitrary shift by updating a level from observed errors [23, 24]. We use the latter idea on the critical level (Theorem 4), where it has a natural feedback signal: whether each unit failed before its replacement. We also measure how badly exchangeability fails across battery batches and N-CMAPSS subsets, and what pilot units cost.

Relation to our earlier evaluation. This article supersedes an evaluation study of the same alarm recipe on the same data [25], which separated prediction accuracy, alarm timing, maintenance cost and transfer and proposed no method. Its numbers are cited as prior results; every number presented as new was produced by the code released with this article.

3. Problem formulation

Units, alarms and notice. A unit observed at rows $j = 1, \dots, n_u$ (cycles $t_1 < \dots < t_{n_u}$) has life L and remaining life $r_j = L - t_j \geq 1$, decreasing in j . A causal signal s_j , here a smoothed RUL estimate, is computed from rows $1, \dots, j$. A threshold trigger with debounce k and warning level H raises an alarm at the first row at which the signal has been at or below H for k consecutive rows. Writing $m_j = \max(s_{j-k+1}, \dots, s_j)$ for $j \geq k$ ($m_j = +\infty$ otherwise) and $\rho_j = \min_{i \leq j} m_i$ for the debounced running minimum, the alarm row is

$$\tau(H) = \min\{j : \rho_j \leq H\} \quad (\tau = \infty \text{ if there is no such row})$$

and the notice is $N(H) = r_{\tau(H)}$ ($N = 0$ without an alarm).

Maintenance consequence. An alarm schedules a replacement ℓ cycles later (the lead time). If $N(H) > \ell$ the planned replacement happens at $t_{\tau(H)} + \ell$ at cost 1; otherwise, or without an alarm, the unit fails at L at cost $\rho > 1$. Units are renewed, so by the renewal-reward theorem the long-run cost per cycle is $E[\text{cost}]/E[\text{use}]$, estimated by total cost over total cycles of use and reported relative to perfect foresight (replacement at $L - 1$, cost 1) [11]. The reliability requirement we target is

$$P(N(H) \leq \ell) \leq \alpha,$$

the probability that the unit fails before the replacement its alarm scheduled.

Window score. The earlier study scored the first alarm by an asymmetric window: with reported estimate $\hat{Y}$ and error $e = \hat{Y} - R$, $R = N(H)$, the alarm is correct if $a \leq e \leq b$, with $[a, b] = [-13, +2]$ cycles for the engines and $[-39, +6]$ cycles for the battery cells.

4. Theory

4.1. The critical level and the notice guarantee

Two elementary properties of the trigger make the notice calibratable.

Lemma 1 (Monotonicity). For $H \leq H'$, $\tau(H') \leq \tau(H)$ and $N(H') \geq N(H)$.

Proof. ρ is non-increasing, so $\{j : \rho_j \leq H\} \subseteq \{j : \rho_j \leq H'\}$; the first element of the larger set is not later, and r_j decreases in j . $\square$

Lemma 2 (Critical level). Let $j_\ell = \max\{j : r_j > \ell\}$ and define the critical level $V(\ell) = \rho_{j_\ell}$ ($V = +\infty$ if no row has $r_j > \ell$). Then for every finite H , $N(H) > \ell \iff H \geq V(\ell)$.

Proof. Because r decreases in j , $N(H) > \ell$ iff the alarm is raised at a row $\tau(H) \leq j_\ell$, iff $\rho_j \leq H$ for some $j \leq j_\ell$, iff $\rho_{j_\ell} \leq H$, because ρ is non-increasing. $\square$

V is the lowest warning level at which the unit's alarm would still have left more than ℓ cycles. It is one number per unit, computed from its signal and its life in a single pass, and it is all that calibration needs.

Theorem 1 (Notice guarantee). Fix the signal map (for example a regressor fitted on units disjoint from those below), the smoothing, the debounce k and the lead time ℓ , and assume that every unit has at least k rows with $r_j > \ell$, so that $V < \infty$. Let the calibration units $1, \dots, n$ and a new unit $n + 1$ be exchangeable, with critical levels $V_1, \dots, V_{n+1}$. For $\alpha \in [1/(n + 1), 1)$ let $q = \lceil (n + 1)(1 - \alpha) \rceil$ and $\hat{H} = V_{(q)}$, the q -th smallest of $V_1, \dots, V_n$. Then

$$P(N_{n+1}(\hat{H}) > \ell) \geq 1 - \alpha, \quad (1)$$

where the probability is over the calibration units and the new unit. If the V_i are almost surely distinct, the probability is also at most $1 - \alpha + 1/(n + 1)$.

Proof. By Lemma 2 the event is $\{V_{n+1} \leq \hat{H}\}$. If $V_{n+1} > V_{(q)}$, at least q calibration values are strictly smaller than V_{n+1} , so the rank of V_{n+1} among all $n + 1$ values (ties broken uniformly at random) is at least $q + 1$. By exchangeability the rank is uniform on $\{1, \dots, n + 1\}$, so $P(V_{n+1} > \hat{H}) \leq (n + 1 - q)/(n + 1) \leq \alpha$. With distinct values, $V_{n+1} \leq V_{(q)}$ iff the rank is at most q , which has probability $q/(n + 1) < 1 - \alpha + 1/(n + 1)$. $\square$

For $\alpha < 1/(n + 1)$ the level is not attainable with n calibration units, and the procedure reports this instead of a level.

Theorem 1 averages over calibration sets. An operator who calibrates once and then runs the alarm on many units faces instead the failure probability given that calibration set, $\pi = P(N_{n+1}(\hat{H}) \leq \ell \mid V_1, \dots, V_n)$, which varies from one calibration set to the next.

Corollary 1 (Training-conditional guarantee). Under the conditions of Theorem 1 with independent and identically distributed units, $P(\pi > x) \leq 1 - \mathcal{B}(x; n + 1 - q, q)$ for every $x \in [0, 1]$, where $\mathcal{B}(\cdot; a, b)$ is the Beta distribution function; if the critical levels are almost surely distinct, π has exactly the Beta($n + 1 - q, q$) distribution. Consequently, the smallest q with $\mathcal{B}(\alpha; n + 1 - q, q) \geq 1 - \delta$ gives $P(\pi \leq \alpha) \geq 1 - \delta$.

Proof. Let G be the distribution function of V and $G^-(u) = \inf\{v : G(v) \geq u\}$. With $U_1, \dots, U_n$ independent uniform, (V_i) has the law of $(G^-(U_i))$, so

$V_{(q)} = G^-(U_{(q)})$ and, by Lemma 2, $\pi = 1 - G(V_{(q)}) \leq 1 - U_{(q)}$ because $G(G^-(u)) \geq u$, with equality when G is continuous. $1 - U_{(q)}$ has the Beta($n + 1 - q, q$) distribution [26]. $\square$

Among m test units sharing one calibration set the number of failures is then Beta-binomial, which predicts how much a failure rate measured on a few dozen units varies around α . With 32 calibration units and $\alpha = 0.10$ ($q = 30$), π exceeds 15% for 12% of calibration sets and exceeds 18.4% for 5%; with 83 units ($q = 76$) the two figures are 5.7% and 15.3%. Being 90% confident that $\pi \leq 0.10$ requires $q = n$ with 32 units (marginal failure probability $1/33 = 3.0\%$) and $q = 79$ with 83 units (6.0%).

Remark 1 (Uncertain lead times). $V(\ell)$ is non-decreasing in ℓ , because j_ℓ is non-increasing and ρ non-increasing. Calibrating at an upper bound $\ell_{\max}$ of the lead time therefore keeps (1) for every lead time up to $\ell_{\max}$. If instead the lead time is random, independent of the units and of known distribution, drawing one lead time ℓ_i per calibration unit and using $V_i(\ell_i)$ keeps the pairs (unit, lead time) exchangeable, and Theorem 1 then bounds the probability of failure before a replacement that arrives after a random lead time.

Corollary 2 (A decision-aware choice keeps the guarantee). Let $q_0 = \lceil (n+1)(1 - \alpha_{\max}) \rceil$ and let $\hat{q} \geq q_0$ be any index chosen from the calibration data, for example to minimise an estimated cost. Then $P(N_{n+1}(V_{(\hat{q})}) > \ell) \geq 1 - \alpha_{\max}$.

Proof. $V_{(\hat{q})} \geq V_{(q_0)}$; by Lemma 1 the event for $V_{(\hat{q})}$ contains the event for $V_{(q_0)}$, whose probability is at least $1 - \alpha_{\max}$ by Theorem 1. $\square$

Proposition 1 (Cost bound and certification floor). (i) Under the conditions of Theorem 1, the expected cost of one renewal cycle of a new unit is at most $1 + (\rho - 1)\alpha$, and its long-run cost rate is at most $(1 + (\rho - 1)\alpha)/E[\text{use}]$. (ii) If the critical levels are almost surely distinct, the alarm at $V_{(q)}$ fails before replacement with probability exactly $(n + 1 - q)/(n + 1)$. Hence no level chosen among the order statistics has a failure probability below $1/(n + 1)$, the expected cost of a renewal cycle is then at least $1 + (\rho - 1)/(n + 1)$, and certifying a failure probability α requires $n \geq 1/\alpha - 1$ calibration units.

Proof. (i) The cost of a cycle is $1 + (\rho - 1)\mathbf{1}\{N \leq \ell\}$; take expectations and apply the renewal-reward theorem. (ii) With distinct values the rank of V_{n+1} is uniform, and the alarm fails iff the rank exceeds q , with probability $(n + 1 - q)/(n + 1)$, which is smallest at $q = n$. $\square$

Proposition 1(ii) is the price of a distribution-free certificate. With 32 calibration units and a failure that costs as much as fifty planned replacements, any order-statistic level carries an expected excess of at least

$49/33 \approx 1.5$ replacement costs per renewal over a policy that never fails. Rules that choose levels beyond the most demanding calibration unit can do better on a given test set, but nothing certifies them.

4.2. Censored calibration data

In a fleet most units are replaced before they fail, so their lives, and hence their critical levels, are not observed. A replaced unit is still informative.

Lemma 3 (Censoring gives an upper bound). If a unit is known to be alive at cycle c ($L > c$), then $V(\ell) \leq U(c)$, where $U(c) = \rho_{j_c}$ with $j_c = \max\{j : t_j \leq c - \ell\}$ ($U = +\infty$ if no such row). $U(c)$ is the critical level the unit would have had if it had failed immediately after c .

Proof. For rows with $t_j \leq c - \ell$, $r_j = L - t_j > c - t_j \geq \ell$, so $j_\ell \geq j_c$ and $V = \rho_{j_\ell} \leq \rho_{j_c}$ because ρ is non-increasing. If the unit had failed immediately after c , its last row with more than ℓ cycles left would be j_c . $\square$

Theorem 2 (Guarantee under arbitrary censoring). Under the conditions of Theorem 1, let each calibration unit report $W_i = V_i$ if its failure was observed and $W_i = U_i(c_i)$ if it was last seen alive at c_i , where the censoring times may depend in any way on the units, their signals and each other. Then $\hat{H}_W = W_{(q)}$ satisfies $P(N_{n+1}(\hat{H}_W) > \ell) \geq 1 - \alpha$.

Proof. By Lemma 3, $W_i \geq V_i$ for every i , so every order statistic satisfies $W_{(q)} \geq V_{(q)}$. By Lemma 1 the event for $\hat{H}_W$ contains the event for $V_{(q)}$, whose probability is at least $1 - \alpha$ by Theorem 1. No assumption on the censoring mechanism is used, because the comparison is pathwise. $\square$

In words: treat every censored unit as if it had failed immediately after it was last seen alive. The guarantee survives any censoring mechanism, including informative censoring, at the price of conservatism that grows with how early units are censored. Dropping censored units (complete-case analysis) keeps the guarantee only when censoring is independent of the unit. Two special cases bracket its value. A unit replaced ℓ cycles after an alarm at level H_0 has $U \leq H_0$; if H_0 lies below the target quantile, such units do not move $\hat{H}_W$ at all, so censoring by an earlier, less conservative alarm is free. Conversely, a unit censored long before its alarm would act as U near the top of the signal's range; if more than about αn units are censored that early, $\hat{H}_W$ reaches the top of the range and the alarm fires at the first possible cycle. The same mechanism means that a conservative level can never be relaxed from units replaced under it. No procedure can avoid this without assumptions on the censoring: an adversarial mechanism could censor exactly the units with the largest critical levels (Section 7.3).

Remark 2 (Complete-case calibration). If the censoring indicators are independent of the units' data, the uncensored calibration units and the new unit remain exchangeable conditionally on which units are censored, so Theorem 1 holds with n replaced by the number of uncensored units. Dropping censored units is then valid and usually far less conservative than Theorem 2; under informative censoring it has no guarantee, in either direction.

4.3. Calibration on all training units by cross-fitting

The split design spends calibration units that could have trained the signal. Cross-fitting uses every training unit for both, at the price of a weaker guarantee. Partition the n training units into K folds, let μ_{-k} be the signal model fitted without fold k by a symmetric algorithm, let V_i be the critical level of unit i under $\mu_{-k(i)}$, and let $\rho_t^{(k)}$ be the debounced running minimum of a new unit under μ_{-k} . The cross-fitted alarm (CNA+) is raised at the first row at which

$$C_t = \#\{i : \rho_t^{(k(i))} \leq V_i\} \geq n+1 - \lceil(1-\alpha)(n+1)\rceil. \quad (2)$$

C_t is non-decreasing in t , so the rule is a trigger.

Theorem 3 (Cross-fitted guarantee). *If the n training units and the new unit are exchangeable and the folds have equal size n/K , then*

$$P(N_{n+1} > \ell) \geq 1 - 2\alpha - \min\left\{\frac{2(1-1/K)}{n/K+1}, \frac{1-K/n}{K+1}\right\}.$$

Proof. Let $V_{n+1}^{(k)}$ be the critical level of the new unit under μ_{-k} . By Lemma 2 applied under each model, $\rho_{j\ell}^{(k)} \leq V_i \iff V_{n+1}^{(k)} \leq V_i$, so the alarm fails to leave more than ℓ cycles iff $C_{j\ell} < n+1 - q$, that is iff $\#\{i : V_{n+1}^{(k(i))} > V_i\} \geq q$ with $q = \lceil(1-\alpha)(n+1)\rceil$. This is the non-coverage event of the CV+ interval of Barber et al. [20] with the nonconformity score “critical level of a unit under a model not fitted on it”. Their proof of the CV+ bound (their Theorem 4) compares scores through a tournament matrix built from leave-fold-out models and uses only the exchangeability of the units and the symmetry of the fitting algorithm, not the form of the score, so it applies verbatim and gives the stated bound. $\square$

The bound is loose for small n (with $n = 80$ and $K = 5$ the finite- K term is about 0.09), and, as for CV+ intervals [20], realised failure rates are usually close to α ; Section 7.4 reports them.

4.4. Online calibration when the population changes

When new units come from a population that differs from the calibration units, none of the guarantees above applies. The failure outcome of each unit, however, is observed whether the unit fails or is replaced,

so the level can be corrected from the new population's own outcomes, as in adaptive conformal inference [23] and conformal quantile tracking [24]. Let units arrive in groups $G_1, G_2, \dots$ of at most g units; all units of group s use the level q_s , and once their outcomes are known

$$q_{s+1} = q_s + \eta \sum_{i \in G_s} (\mathbf{1}\{N_i(q_s) \leq \ell\} - \alpha), \quad \eta > 0. \quad (3)$$

Theorem 4 (Long-run failure rate for any sequence of units). *Suppose the critical levels of all units lie in an interval $[a, b]$ of width B and $q_1 \in [a, b]$. Then for every sequence of units, without any exchangeability or distributional assumption, after T units*

$$\left| \frac{1}{T} \sum_{i=1}^T \mathbf{1}\{N_i \leq \ell\} - \alpha \right| \leq \frac{B + g\eta}{\eta T}.$$

Proof. By Lemma 2, unit i of group s fails iff $V_i > q_s$. We show by induction that $q_s \in [a - g\alpha\eta, b + g(1-\alpha)\eta]$. If $q_s > b$, no unit of the group fails and $q_{s+1} = q_s - \eta\alpha|G_s|$, which lies in $[q_s - g\alpha\eta, q_s]$. If $q_s < a$, every unit fails and $q_{s+1} = q_s + \eta(1-\alpha)|G_s| \in (q_s, a + g(1-\alpha)\eta]$. If $q_s \in [a, b]$, then $q_{s+1} \in [a - g\alpha\eta, b + g(1-\alpha)\eta]$. In every case the claim carries over. Summing (3), $\sum_{i \leq T} (\mathbf{1}\{N_i \leq \ell\} - \alpha) = (q_{S+1} - q_1)/\eta$, and $|q_{S+1} - q_1| \leq B + g\eta$ because both lie in the interval above and $q_1 \in [a, b]$. $\square$

Theorem 4 trades the per-unit guarantee of Theorem 1 for a deterministic statement about the realised failure rate of the sequence, which holds whatever the population does. The price is that early units of a shifted population may fail before the level has moved, the bound is loose for short sequences, and in a fleet the outcomes of units operated concurrently arrive late, which the group size g accounts for. The step η trades speed of adaptation against the variability of the level.

4.5. What a window score and a per-cycle interval certify

Proposition 2 (Window score of an estimate versus a constant). *Let $R \in \mathbb{Z}$ be the notice at the alarm, $\hat{Y}$ the reported estimate, $[a, b]$ the acceptance window for $e = \hat{Y} - R$, c any constant and $D = \hat{Y} - c$. With the half-open intervals $I_b = [c - b, c - b + D]$ and $I_a = (c - a, c - a + D]$ for $D \geq 0$ (mirrored for $D < 0$): (i) $|\mathbf{1}\{a \leq \hat{Y} - R \leq b\} - \mathbf{1}\{a \leq c - R \leq b\}| \leq \mathbf{1}\{R \in I_a\} + \mathbf{1}\{R \in I_b\}$; (ii) if, given $\hat{Y}$, the probability mass function of R is at most π_a within Δ of $c - a$ and at most π_b within Δ of $c - b$, and $|D| \leq \Delta$, then the window scores satisfy $|S_{\text{est}} - S_c| \leq (\pi_a + \pi_b)E[|D|]$; (iii) if the reported estimate is the triggering signal, $\hat{Y} = s_\tau$, and $\tau > k$, then $H - k\delta^+ \leq \hat{Y} \leq H$ with $\delta^+ = \max(0, \max_{\tau-k < i \leq \tau} (s_{i-1} - s_i))$.*

Proof. (i) For $D \geq 0$ the estimate accepts $R \in [c - b + D, c - a + D]$ and the constant $R \in [c - b, c - a]$. R

accepted only by the estimate satisfies $c - a < R \leq c - a + D$; accepted only by the constant, $c - b \leq R < c - b + D$. The case $D < 0$ is symmetric, and the argument holds also when the two sets are disjoint. (ii) Take expectations of (i); a half-open interval of length $|D|$ contains at most $\lceil |D| \rceil$ integers. (iii) $\hat{Y} = s_\tau \leq m_\tau \leq H$. As the alarm was not raised at $\tau - 1 \geq k$, some row $i_0 \in [\tau - k, \tau - 1]$ has $s_{i_0} > H$, and $s_\tau = s_{i_0} - \sum_{i=i_0+1}^\tau (s_{i-1} - s_i) > H - k\delta^+$. $\square$

For a threshold-crossing alarm the reported estimate is pinned to a band of width $k\delta^+$ below the level, so a window score is, up to the edge term of (ii), the probability that the notice falls in a band $[c - b, c - a]$. It can be high while the notice is too short for the lead time, so it cannot certify the event of Theorem 1. Nor can a per-cycle conformal lower bound $\text{LB}_t = s_t - q_\alpha$: its coverage $P(r_t \geq \text{LB}_t) \geq 1 - \alpha$ holds for a row drawn independently of the data, whereas the alarm “act when $\text{LB}_t \leq \ell + 1$ ” is read at the first row at which the bound crosses a level, where an optimistic error is most likely. That alarm is a threshold trigger at level $\ell + 1 + q_\alpha$ on the same signal, so it differs from CNA only in how the level is calibrated, which makes it the natural comparator (Section 7.1).

5. Method: the conformal notice alarm

Algorithm 1 Conformal notice alarm (CNA) and its variants

Input: causal signal map; debounce k ; lead time ℓ ; level α ; calibration units disjoint from those used to fit the signal.

Calibrate. For each calibration unit i : compute its debounced running minimum $\rho^{(i)}$ and its critical level $V_i = \rho_{j_\ell}^{(i)}$ (Lemma 2). If the unit was replaced or last seen alive at cycle c_i , use $U_i(c_i)$ instead (Lemma 3). Set $\hat{H} = V_{(\lceil (n+1)(1-\alpha) \rceil)}$.

Operate. Raise the alarm at the first row at which the signal has been at or below $\hat{H}$ for k consecutive rows; schedule the replacement ℓ cycles later.

CNA-DA. Given a cost ratio ρ and a ceiling $\alpha_{\max}$, use $V_{(\hat{q})}$ with $\hat{q} = \arg \min_{q \geq q_0} [1 + (\rho - 1) \frac{n+1-q}{n+1}] / \text{use}(V_{(q)})$, $q_0 = \lceil (n+1)(1 - \alpha_{\max}) \rceil$, with the mean cycles of use estimated on the calibration units.

CNA+. Replace the split by K -fold cross-fitting and raise the alarm by rule (2).

Online (new population). Start from $\hat{H}$ of the old population; after each group of new units, update the level by (3) with the observed outcomes (Theorem 4).

which sees no failure among calibration units at the edge of the feasible region and was fragile in the earlier study [25]. By Corollary 2 the failure probability of CNA-DA is at most $\alpha_{\max}$ whatever ρ is.

Calibration costs one pass over the calibration series (Fig. 1). The numerator of the CNA-DA objective replaces the empirical failure rate by its conformal value, which is never zero: with $q = n$ it is $1/(n+1)$. This is the difference from plain empirical cost minimisation,

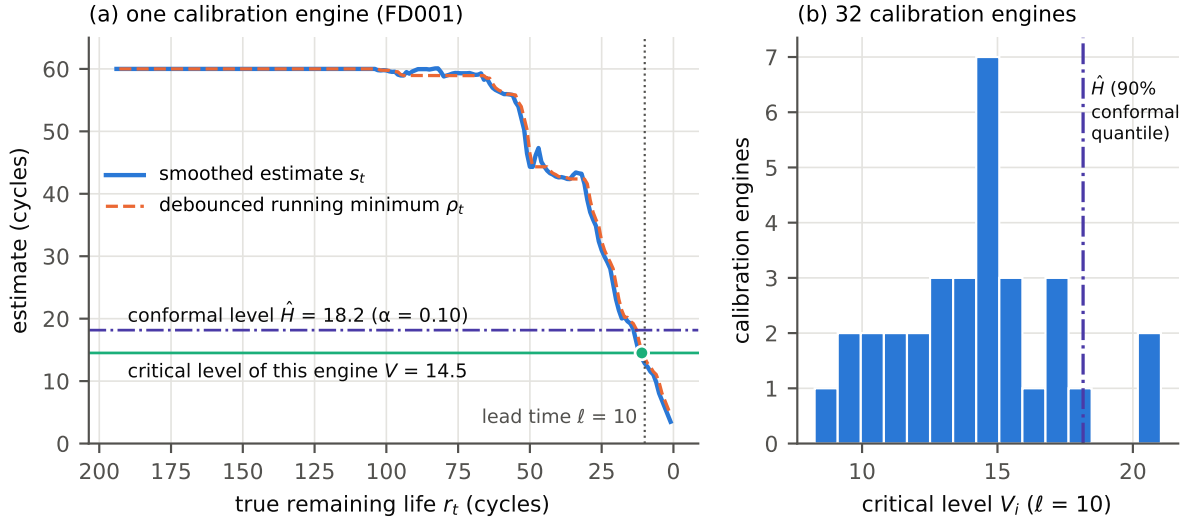


Fig. 1: The conformal notice alarm on FD001 (fold 1, lead time 10 cycles). (a) The smoothed estimate of one calibration engine and its debounced running minimum ρ_t ; the critical level V is the value of ρ at the last cycle with more than ℓ cycles left (dot). (b) Critical levels of the 32 calibration engines and the conformal level $\hat{H}$ at $\alpha = 0.10$.

6. Experimental design

The protocol (hypotheses H1–H5, datasets, policies, metrics and decision rules) was written and hashed with SHA-256 before the main experiments. Four amendments were hashed before their own experiments, each after the earlier results were known: H6–H8 (censoring, cross-fitting and the setting of Kamariotis et al.), H9 (comparators with the same information as the cross-fitted alarm), H10 (online calibration under batch shift) and H11–H13 (other learners and alarm settings). The hashes are not registry preregistrations; they show that the files have not changed since, and their timing rests on the repository history. The code was smoke-tested on one fold before each hash. One analysis was added afterwards and is labelled exploratory: the check of Corollary 1 on fold-level failure counts.

Data. We use the 218 PHM08 training engines (the development dataset of the earlier study), the C-MAPSS FD001–FD004 training sets (100, 260, 100 and 249 engines) [1], N-CMAPSS reduced to one row per flight (99 engines in nine subsets, six temperature and speed channels) [27], and 123 lithium-ion cells of Severson et al. [28] in three batches (36, 43 and 44 cells; end of life at 80% of nominal capacity), aggregated into blocks of 3 cycles fixed a priori. The engine datasets are simulations from one simulator family; the battery data are laboratory measurements. The 17 PRONOSTIA bearings [29] of the earlier study are excluded: with at most about six calibration bearings per split, $\alpha \leq 0.10$ is not attainable.

Splits and information. Outer splits are five-fold cross-validation by unit (seed 2026), in which test units are exchangeable with training units, and, for transfer,

leave-one-batch-out on the battery cells and leave-one-subset-out on N-CMAPSS. Inside each outer training set, 60% of the units (proper training) fit the normalisation and the regressors and 40% (calibration) are used for every selection, tuning and calibration step of every policy; CNA+ instead cross-fits all training units. Outer test units are scored once. The proper-training/calibration split is repeated three times (repetition 0 is primary). Disjointness of the unit sets is asserted in the code; features at row t use rows $\leq t$ only (tested by truncation); no test life enters any setting.

Policies. The signal is the regressor of the earlier study: histogram gradient boosting with absolute-error loss (500 iterations, learning rate 0.05, 31 leaves, at least 40 samples per leaf) on causal features of regime-normalised sensors (early-life baseline, exponentially weighted means, local slopes, age), target $\min(r, c)$ [30]. Table 1 lists the policies. CNA uses settings fixed a priori (cap 60 cycles, battery 180; span 3; debounce 2), so that no selection touches the calibration units before the conformal step. Costs use lead times $\ell \in \{0, 5, 10, 15\}$ cycles (battery $\{0, 15, 30, 45\}$) and cost ratios $\rho \in \{5, 10, 20, 50\}$; the primary lead time is 10 cycles (battery 30). The cost ratios are illustrative.

Metrics, tests and decision rules. The primary metric is the *failure-before-replacement rate* on test units, with Clopper–Pearson intervals [31] and a one-sided exact binomial test of rate $> \alpha$, Holm-adjusted over datasets [32]. Notice at the alarm and wasted life measure what the guarantee costs. Cost rates are compared by paired bootstrap ratios over test units (2,000 resamples); a policy is called cheaper than another on a dataset if it is significantly cheaper in at least 8 of the 16 cost cells and significantly dearer in none. Window

Table 1

Policies compared. All settings of learned policies are selected on calibration units; regressors are fitted on proper-training units (CNA+: cross-fitted on all training units).

Policy	Signal and trigger	Selected on calibration units	Guarantee
CNA ($\alpha = 0.05, 0.10, 0.20$)	GBM estimate (cap 60; battery 180 cycles), span 3, debounce 2, fixed a priori	$\hat{H}$ = conformal quantile of critical levels	$P(\text{fail}) \leq \alpha$ (Thm 1)
CNA-DA	As CNA	Order statistic minimising the cost bound, $\alpha_{\max} = 0.10$	≤ 0.10 (Cor. 2)
CNA+ / CNA+-DA	As CNA, five cross-fitted models	Count threshold of rule (2)	$\leq 2\alpha + O(K/n)$ (Thm 3)
Online CNA (new population)	As CNA	Level updated after each new unit by (3) from the old population's $\hat{H}$	Long-run rate (Thm 4)
LB-all / LB-near	As CNA; alarm when $s_t - q_\alpha \leq \ell + 1$	Conformal quantile of per-row residuals (all rows / rows with $r \leq \text{cap}$)	Per-row only
STW, window-tuned	GBM estimate (cap 40/60/90), span, debounce, shift	Most correct first warnings per H ; H by calibration cost	None
STW, cost-tuned (plain, 1-SE)	GBM estimate, span, debounce, threshold (2,160 settings)	Lowest calibration cost; 1-SE: largest notice within one bootstrap SE	None
Kamariotis P1 (1/ ρ , optimised)	$P(r \leq \ell + \Delta t \mid \text{estimate bin})$ from calibration rows (P-res)	Threshold 1/ ρ , or optimised on calibration cost	None
Age replacement	Age	Replacement age on training lives	None
Capacity trend (battery)	Discharge capacity extrapolated to 0.88 Ah	As the STW and CNA counterparts, on all training cells	CNA version: Thm 1
CF versions (STW cost-tuned plain and 1-SE; P1 optimised and 1/ ρ)	Average of the five fold models of CNA+	Out-of-fold predictions of all training units	None

scores of estimate and constant are compared by the Tango score interval for paired proportions [33] and two one-sided tests [34] with a margin of 5 points. Intervals cover the sampling of test units given fitted policies; the repetitions show the variability from the split.

7. Results

All numbers are on outer test units. Calibration sets had 32 units (FD001, FD003, N-CMAPSS), 39 (battery), 70 (PHM08), 80 (FD004) and 83 (FD002) under cross-validation. Unless stated, rates are averaged over the three repetitions of the proper-training/calibration split, and tests use repetition 0, in which every unit is tested once.

7.1. The guarantee holds; interval calibration does not deliver it

Table 2 and Fig. 2 give the failure-before-replacement rate at the primary lead time. With $\alpha = 0.10$ the conformal notice alarm failed on 6.0–11.7% of test units across the seven datasets, and with $\alpha = 0.05$ and 0.20 on 1.3–5.1% and 14.3–20.6%. Pooled over the seven datasets, four lead times and three repetitions (13,788 unit–cell evaluations), the rates were 3.8, 9.3 and 19.4% for nominal 5, 10 and 20%. In repetition 0 no exact binomial test of rate > 0.10 was significant after Holm's correction (smallest adjusted $p = 0.84$), so H1 is supported on all seven datasets. Rates slightly above 10% (FD003 11.7%, N-CMAPSS 11.1%, battery 10.6%, FD004 10.4%) are within sampling error; for

distinct critical levels the expected rate is $(n+1-q)/(n+1)$, 9.1% with 32 calibration units and 10.0% with 39.

The interval-calibrated alarm on the same signal and trigger did not deliver its nominal level. Calibrated with residuals over all rows (LB-all), the lower-bound alarm at nominal 10% failed on 18.2% (N-CMAPSS) to 69.6% (battery) of units, 46.6% pooled, with $p < 0.005$ on every dataset (H2 supported, 7 of 7). Rows far from failure, where the capped target makes residuals small, dominate the pooled residual quantile, so the bound is tight far from failure and loose at the stopping time. Restricting the residuals to rows with $r_t \leq c$ (LB-near) brought the rate close to nominal on six datasets (4.0–8.6%) but not on N-CMAPSS (15.2%, and 17.8% at the longest lead time), and its level depends on how the evaluator chooses the rows. The guarantee costs notice: at $\alpha = 0.10$ CNA alarms left a median of 15–17 engine cycles at a lead time of 10, against 11–12 for LB-all; on the battery cells 43 cycles at a lead time of 30.

Corollary 1 predicts more than the mean. Across the 105 calibration sets at the primary lead time (seven datasets, five folds, three repetitions), the number of failures among the test units of a fold followed the Beta-binomial law that the corollary implies: the randomised probability integral transforms were consistent with uniformity (Kolmogorov–Smirnov $p = 0.93$ at $\alpha = 0.10$; 0.53 and 0.70 at 0.05 and 0.20), the variance of the fold-level failure rate at $\alpha = 0.10$ was 0.0049 against 0.0048 predicted, and 103 of the 105 rates lay inside the 90% predictive band (Fig. 3). For folds of 20–25 units the band reaches 25–26%: a valid 10% alarm can show more than twice its nominal

Table 2

Failure before the planned replacement at the primary lead time (10 cycles; battery 30 cycles), five-fold cross-validation, mean of three repetitions. Second column: repetition 0 for CNA at $\alpha = 0.10$ (failures; rate; Clopper–Pearson 95 % interval). Last column: median notice at the alarm (cycles), CNA 0.10 / LB-all 0.10.

Dataset (test units)	CNA 0.10, rep. 0	CNA 0.05	CNA 0.10	CNA 0.20	CNA-DA	LB-all 0.10	LB-near 0.10	Notice
PHM08 (218)	18 (0.083; 0.050–0.127)	0.034	0.086	0.177	0.009	0.446	0.080	16 / 11
FD001 (100)	7 (0.070; 0.029–0.139)	0.013	0.060	0.143	0.013	0.333	0.043	17 / 12
FD002 (260)	20 (0.077; 0.048–0.116)	0.042	0.088	0.191	0.008	0.381	0.040	15 / 12
FD003 (100)	14 (0.140; 0.079–0.224)	0.027	0.117	0.197	0.027	0.480	0.063	16 / 11
FD004 (249)	31 (0.124; 0.086–0.172)	0.051	0.104	0.206	0.011	0.466	0.086	16 / 11
N-CMAPSS (99)	11 (0.111; 0.057–0.190)	0.037	0.111	0.178	0.057	0.182	0.152	23 / 19
Battery (123)	15 (0.122; 0.070–0.193)	0.033	0.106	0.198	0.011	0.696	0.070	43 / 28

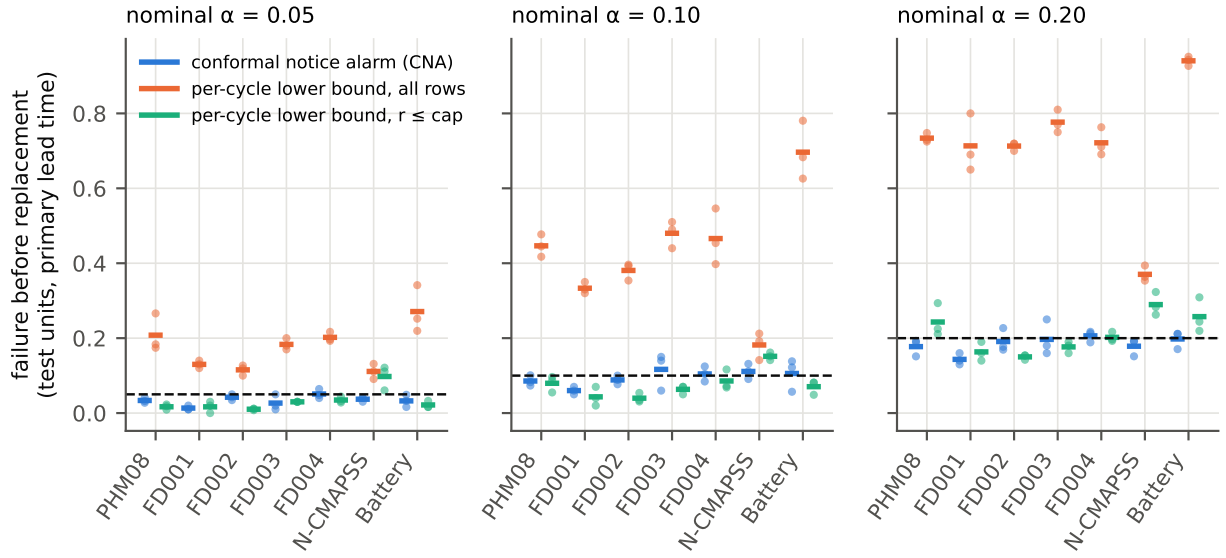


Fig. 2: Failure before the planned replacement at the primary lead time (10 cycles; battery 30) for the conformal notice alarm (CNA) and per-cycle split-conformal lower-bound alarms on the same signal and trigger, at three nominal levels (dashed). Dots: the three repetitions; bars: their mean. Five-fold cross-validation.

rate on a test of that size. This check was not in the protocols, and folds of one repetition share calibration units, so the p -values are descriptive.

7.2. Other learners and alarm settings

The fourth amendment repeated the validity analysis (repetition 0, same units and splits) with three further learners, ridge regression and a random forest on the causal features and a multilayer perceptron on the raw normalised sensors of the last 30 rows, and with 27 settings of the GBM signal (cap, smoothing span and debounce, three values each). The GBM reference reproduced the main run's decisions exactly on all seven datasets. With every learner the conformal notice alarm kept its level (Fig. 4a): at $\alpha = 0.10$ and the primary lead time the three new learners failed on 4.0–17.0 % of test units on individual datasets and on 9.3–9.9 % pooled over datasets, and no exact test exceeded 10 % after Holm's correction (smallest adjusted $p = 0.43$; H11 supported). Pooled over datasets and lead

times, all four learners gave 3.9–4.2, 9.5–10.0 and 18.5–19.6 % at nominal 5, 10 and 20 %. The per-cycle lower bound calibrated over all rows failed on 20–70 % of units with the random forest and 26–71 % with ridge regression, significantly above 10 % on all seven datasets, and on 6–38 % with the perceptron, significantly on five (H12 supported). Restricting its residuals to rows near failure, which nearly repaired it for GBM, did not repair it for ridge regression (17–33 %) and only partly for the random forest (up to 17 %): how well that repair works depends on the learner as well as on the choice of rows. A weaker signal cost CNA notice rather than reliability: with ridge regression the median notice at the primary lead time was 18–21 cycles on PHM08 and C-MAPSS against 15.5–16 with GBM (on N-CMAPSS 23 against 25), and with the perceptron 64 battery cycles against 42. Across the 27 settings of the GBM signal the pooled failure rate stayed between 8.7 and 11.1 % (H13 supported) and the mean notice between 17.4 and 18.1 engine cycles

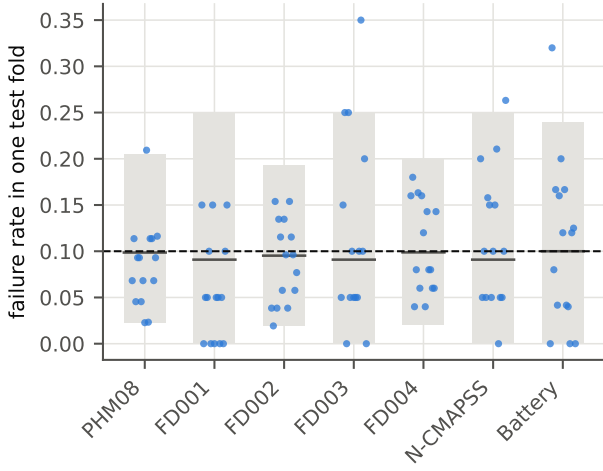


Fig. 3: Failure rate of CNA ($\alpha = 0.10$, primary lead time) in each of the 105 test folds (five folds, three repetitions per dataset) against the Beta-binomial law of Corollary 1: 90% predictive band for the median fold size (grey) and its mean $(n + 1 - q)/(n + 1)$ (line).

(battery 41–48): once the critical level is calibrated, the a-priori choice of these settings hardly matters for either.

7.3. Censored calibration data

Table 3 and Fig. 5 apply the three censoring mechanisms of the first amendment to the calibration units ($\alpha = 0.10$, primary lead time, repetition 0, 50 draws per split). Treating every censored unit as failed immediately after it was last seen alive kept the failure rate at or below its nominal level under every mechanism and censoring fraction on every dataset (H6 supported, 49 of 49 cells), as Theorem 2 requires. Whether it was *useful* depended entirely on when units were censored.

When the calibration history came from an earlier alarm at the median critical level, about half of the calibration units (48–51%) were censored, and the censored-as-failure level coincided with the level computed from complete data in every draw at $\alpha = 0.10$ (and in 99.3–100% of draws at $\alpha = 0.20$): failure rates and notice were identical to the uncensored analysis on all seven datasets. Censoring of this kind, which is what a fleet operated under an alarm produces, cost nothing. Dropping the censored units instead was over-conservative here (failure 2.8–5.5% against 7.0–14.0% uncensored), because the earlier alarm had censored precisely the units with low critical levels.

Under random and administrative censoring, which remove units at arbitrary points of their lives, the guarantee survived but its price was prohibitive: with a quarter of the calibration units censored, the upper bounds of the early-censored units pushed the level to the top of the signal's range, and the alarm fired

near the start of life (mean notice at planned replacements 34–751 cycles against 17–46 without censoring). Complete-case calibration stayed close to the uncensored analysis up to half of the units censored (failure 4.5–12.2%), but it carries a guarantee only when censoring is independent of the units (Remark 2), which the administrative mechanism violates by design; with three quarters censored, the 8 remaining calibration units of FD001, FD003 and N-CMAPSS could not certify $\alpha = 0.10$ in most draws. The practical reading is that censored fleet data are usable when units are replaced late in life, by an alarm or for reasons unrelated to the units, and that a unit replaced long before an alarm would have acted carries almost no information about how late it could have been replaced.

7.4. Cost: the price of the certificate, and how cross-fitting lowers it

Split design. The decision-aware split alarm (CNA-DA, $\alpha_{\max} = 0.10$) kept its failure rate at or below $\alpha_{\max}$ in every lead-time and cost-ratio cell with a positive lead time (largest 6.1%; H4b supported) and failed on 1.8% of units over all cells and repetitions. It was cheaper than age replacement on PHM08, FD002, FD004 and the battery cells (12–16 of 16 cells; never significantly dearer anywhere), than the probability-threshold policy with threshold $1/\rho$ on PHM08, FD002 and FD004, and than the optimised probability-threshold policy on PHM08, FD001 and the battery cells (8 of 16 cells each). It was not the cheapest policy (H4 not supported): a cost-tuned alarm with a one-standard-error rule was cheaper in 8 of 16 cells on FD001 and 11 on FD003, and the capacity trend was the cheapest policy on the battery cells. Proposition 1(ii) explains most of the gap: with 32 calibration units no order-statistic level fails less often than 3%, and at $\rho = 50$ each such failure costs as much as 49 planned replacements; the cost ratios against the one-standard-error rule grew with ρ and were largest on the two datasets with 32 calibration units (FD001 and FD003; median ratio 1.37 and 1.45) and smallest on those with 70–83 (PHM08 and FD002; 1.00 and 1.04).

Cross-fitting. Calibrating on all training units by cross-fitting attacks that floor directly. The cross-fitted alarm CNA+ at $\alpha = 0.10$ failed on 3.8–10.1% of test units at the primary lead time (repetition 0; H7a supported; 4.3–10.1% averaged over the three repetitions), within the nominal level on every dataset although Theorem 3 guarantees only 2α plus a finite- K term, and it gave the same notice as the split alarm (median 15–22 engine cycles; battery 42). Its decision-aware version was cheaper than the split CNA-DA on FD001, FD003 and FD004 (8–12 of 16 cells, never dearer) and mixed elsewhere, so the protocol's prediction of at least four datasets (H7b) was not met. Its mean excess cost over the cheapest policy in each

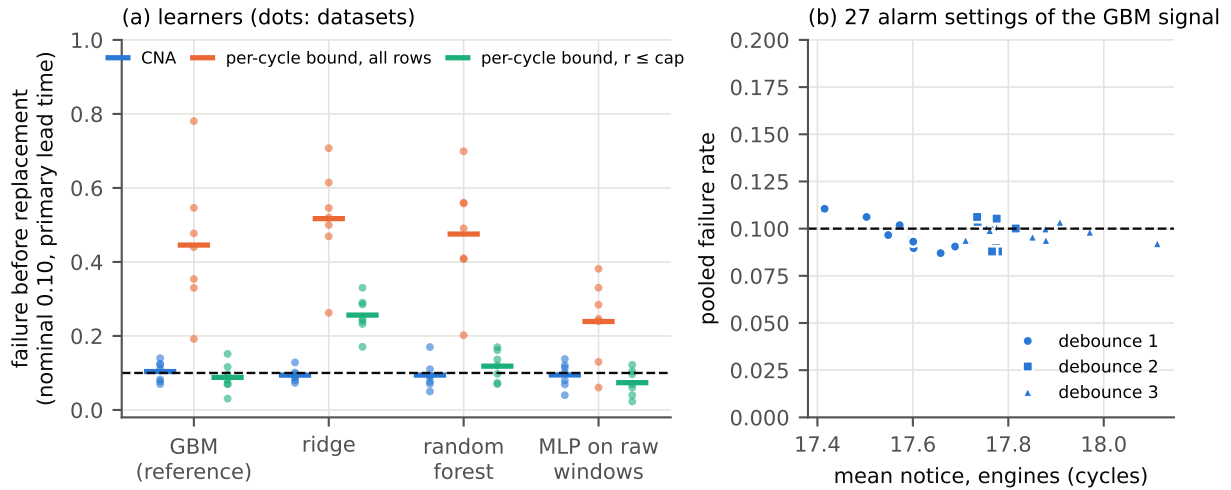


Fig. 4: Other learners and alarm settings (repetition 0, primary lead time). (a) Failure before replacement at nominal 0.10 for CNA and the per-cycle lower-bound alarms built on four learners fitted to the same units; dots: datasets, bars: their mean. (b) Pooled failure rate of CNA at $\alpha = 0.10$ against the mean notice on the engine datasets for 27 settings of the GBM signal (cap $\times$ smoothing span $\times$ debounce).

Table 3

Censored calibration units ($\alpha = 0.10$, primary lead time, repetition 0, mean over 50 censoring draws and five folds): failure before replacement and, in brackets, mean notice at planned replacements (cycles). Policy: history under an earlier alarm at the median critical level (48–51 % censored).

Mechanism	Censored	Method	PHM08	FD001	FD002	FD003	FD004	N-CMAPSS	Battery
None	0	complete data	0.083 (17)	0.070 (18)	0.077 (17)	0.140 (17)	0.124 (20)	0.111 (28)	0.122 (46)
Policy	$\approx 50\%$	censored as failure	0.083 (17)	0.070 (18)	0.077 (17)	0.140 (17)	0.124 (20)	0.111 (28)	0.122 (46)
		complete case	0.034 (18)	0.028 (25)	0.032 (19)	0.055 (29)	0.054 (23)	0.046 (33)	0.035 (66)
Random	25 %	censored as failure	0.000 (175)	0.000 (166)	0.000 (169)	0.001 (180)	0.000 (196)	0.005 (47)	0.000 (654)
		complete case	0.083 (17)	0.065 (18)	0.079 (17)	0.101 (18)	0.116 (20)	0.098 (28)	0.107 (48)
	50 %	complete case	0.072 (17)	0.054 (20)	0.076 (17)	0.093 (21)	0.108 (21)	0.082 (31)	0.094 (52)
Administrative	25 %	censored as failure	0.000 (72)	0.000 (99)	0.000 (77)	0.000 (140)	0.000 (150)	0.036 (34)	0.000 (751)
		complete case	0.081 (17)	0.074 (18)	0.079 (17)	0.098 (18)	0.122 (20)	0.088 (29)	0.114 (47)
	50 %	complete case	0.066 (17)	0.063 (19)	0.071 (17)	0.077 (20)	0.109 (21)	0.045 (34)	0.102 (52)

cell fell to 0.4–19.5 % (from 8.5–85.6 % for the split version; Table 4, Fig. 6b).

Equal information. The comparison above favours CNA+, which calibrates on all training units while the uncertified policies were tuned on the 60/40 split. The third protocol therefore rebuilt the comparators on exactly the same five fold models: selected on the out-of-fold predictions of all training units and applied through the average of the five models (CF in Table 4). With equal information the certified alarm cost about as much as the best uncertified rules (H9). Its mean excess over the cheapest of the six policies in this comparison was 1.0–14.3 % across the seven datasets, against 0.6–11.0 % for the one-standard-error cost-tuned alarm and 1.8–13.5 % for policy 1 of Kamaritis et al. with an optimised threshold. Averaged over the three repetitions (descriptive) the three ranges were 1.9–14.6, 0.9–11.7 and 1.9–11.8 %, and the excess of CNA+-DA moved by up to 12 points between repetitions (N-CMAPSS), so differences of a few points among these three policies are within the variability

of the split. By the protocol's cell-count rule CNA+-DA was never dearer than either: it was cheaper than the one-standard-error alarm on the battery cells (16 of 16 cells) and on FD004 (8 of 16) and mixed on the other five datasets, and mixed against the optimised probability-threshold policy on all seven. It was cheaper than policy 1 with the heuristic threshold $1/\rho$ on PHM08, FD002 and FD004. Plain cost tuning won many cells by small margins but lost heavily in a few, because it occasionally chose a setting at the edge of the feasible region: its largest excess in a cell was 97 % on FD003, against 45 % for CNA+-DA. Where CNA+-DA did lose, it lost at $\rho = 50$: with about 80 calibration units its failure probability cannot fall below about 1.2 % (Proposition 1), and the one-standard-error alarm, which can choose levels beyond the most demanding calibration unit, had no test failure in those cells. The one-standard-error and optimised-threshold comparators failed on 0–1 % of test units at the primary lead time and $\rho = 10$ (all three repetitions), the heuristic threshold on 3.3–13.1 %; the

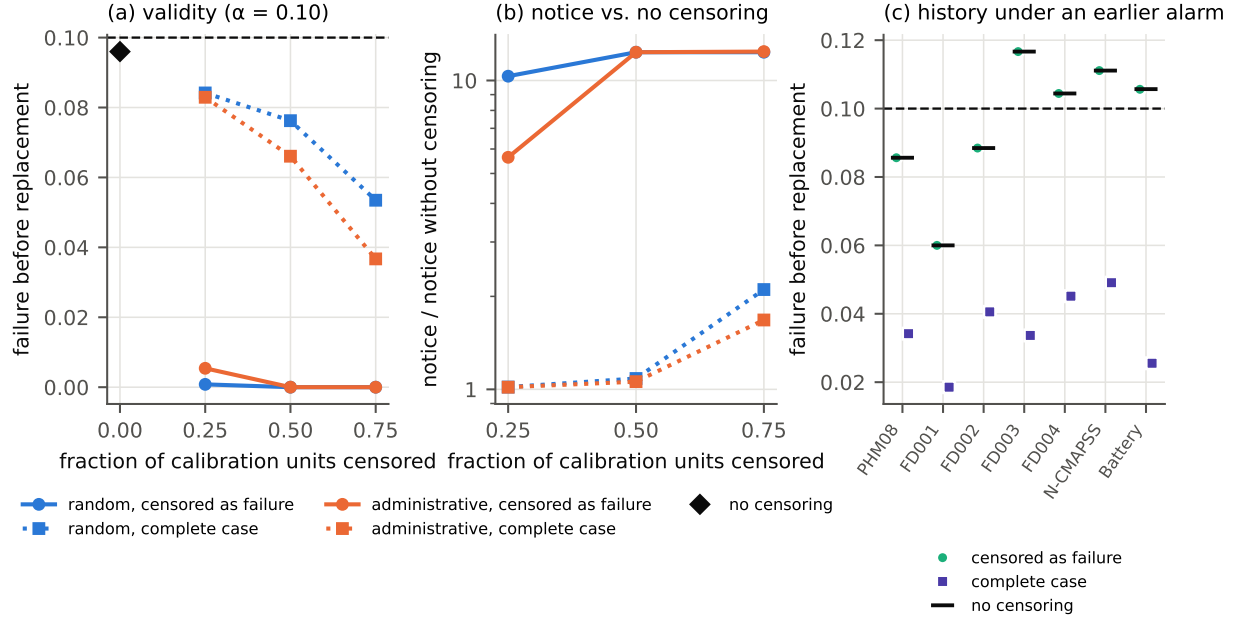


Fig. 5: Censored calibration units ($\alpha = 0.10$, primary lead time, repetition 0). (a) Failure before replacement against the fraction of calibration units censored, mean over datasets. (b) Notice relative to the uncensored analysis (log scale). (c) Calibration history produced by an earlier alarm: censored-as-failure reproduces the uncensored result (bars) exactly; complete-case analysis is over-conservative.

Table 4

Mean excess cost over the cheapest policy in each of the 16 lead-time and cost-ratio cells (%; cross-validation, repetition 0). Rows 1–3 are certified; the others are not. “CF”: selected on cross-fitted out-of-fold predictions of all training units and applied through the average of the same five fold models as CNA+ (equal information with CNA+); the split policies use 60 % of the training units for fitting and 40 % for selection.

Policy	PHM08	FD001	FD002	FD003	FD004	N-CMAPSS	Battery
CNA-DA, split (certified)	8.5	64.3	13.6	85.6	19.9	50.4	30.4
CNA+-DA, cross-fitted (certified, 2α bound)	6.9	9.0	0.4	13.1	1.9	5.6	19.5
CNA-DA on capacity trend (certified)	–	–	–	–	–	–	18.8
Cost-tuned STW, 1-SE, split	0.9	0.7	4.6	6.2	2.8	51.2	6.1
Window-tuned STW, split	2.0	36.3	4.1	38.6	13.9	29.3	29.8
Policy 1, optimised, split	4.4	51.9	8.3	42.7	16.8	12.9	27.1
Capacity trend, 1-SE	–	–	–	–	–	–	1.9
Age replacement	80.6	72.9	57.6	89.5	80.8	18.8	179.7
<i>Equal information with CNA+ (same five fold models; excess over the cheapest of these six policies; policy 1 with threshold $1/\rho$ not shown)</i>							
CNA+-DA (certified)	8.9	14.3	1.0	12.8	3.8	13.0	1.5
Cost-tuned STW, 1-SE, CF	0.6	1.0	1.1	0.8	1.2	11.0	7.6
Cost-tuned STW, plain, CF	6.0	18.6	6.7	21.7	6.7	19.6	1.3
Policy 1, optimised, CF	4.3	9.9	1.8	13.5	3.4	10.4	2.0
Age replacement	84.2	81.0	58.5	89.2	84.6	26.8	132.5

former achieved low failure rates on these units, but without a guarantee for the next one.

7.5. The setting and metric of Kamariotis et al.

On their decision grid (decisions every 10 cycles, no lead time) and with their metric M , the split CNA-DA reached $M = 11.9$ – 34.1 % at $c_p/c_c = 0.1$ on PHM08 and FD001–FD004 (Table 5, Fig. 6c). It was better than their policy 1 with an optimised threshold on FD001

(-1.3 points, paired 95 % interval -2.0 to -0.7) and indistinguishable from it on the other four datasets, better than policy 1 with the heuristic threshold c_p/c_c on FD004, and better than age replacement on all five. The cost-tuned alarm with the one-standard-error rule had the lowest M on every dataset (4.4–18.1 %) and was significantly better than CNA-DA on FD002 and FD004. At $c_p/c_c = 0.02$ the floor of Proposition 1 dominates: with 32–83 calibration units, a certified failure probability of 1–3 % costs as much as 50 preventive

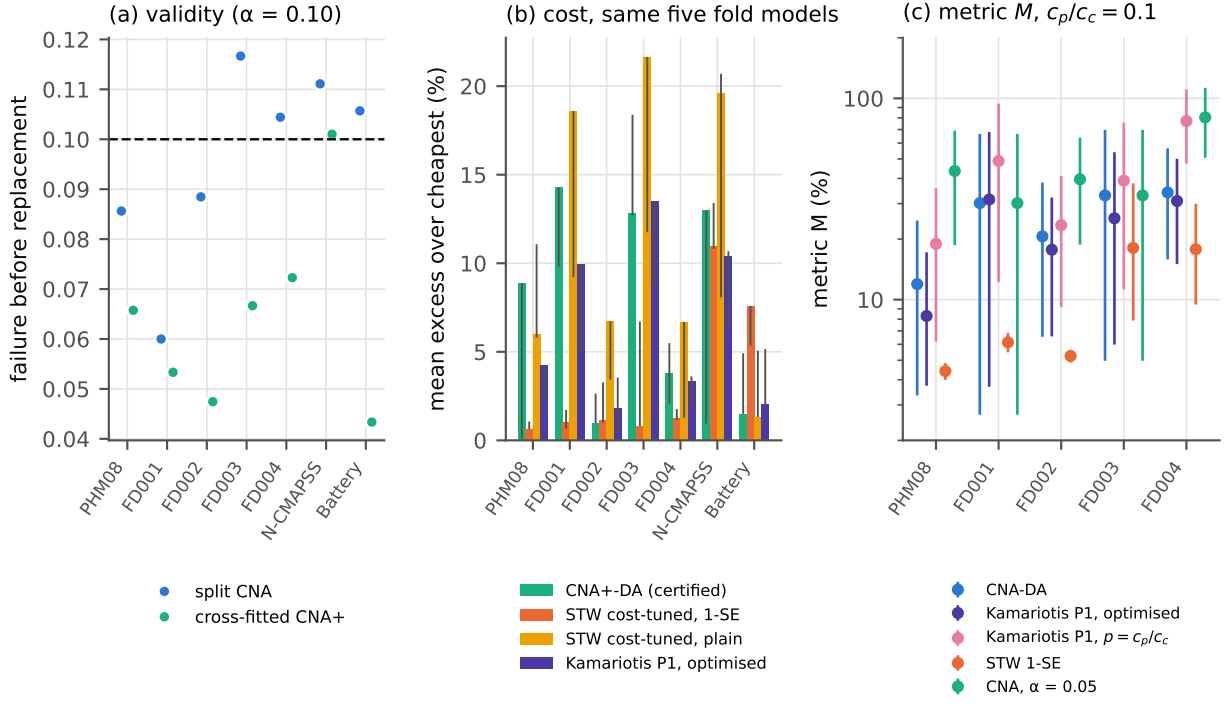


Fig. 6: (a) Failure before replacement at $\alpha = 0.10$ and the primary lead time for the split (CNA) and cross-fitted (CNA+) alarms. (b) Mean excess cost over the cheapest policy in each of the 16 cells for the cross-fitted certified alarm and the uncertified policies built on the same five fold models (equal information); bars: repetition 0, lines: range over the three repetitions. (c) Metric M in the setting of Kamariotis et al. at $c_p/c_c = 0.1$ with paired-bootstrap 95 % intervals (log scale).

Table 5

Metric M (%) in the setting of Kamariotis et al. ($\Delta T = 10$ cycles, no lead time, $c_p/c_c = 0.1$; five-fold cross-validation, repetition 0) and, in brackets, test failures.

Policy	PHM08	FD001	FD002	FD003	FD004
CNA-DA (certified, $\alpha_{\max} = 0.10$)	11.9 (2)	30.2 (3)	20.6 (5)	33.0 (3)	34.1 (7)
CNA, $\alpha = 0.05$ (certified)	43.6 (10)	30.2 (3)	39.6 (11)	33.0 (3)	80.5 (21)
Policy 1, optimised threshold	8.3 (1)	31.4 (3)	17.7 (4)	25.4 (2)	30.9 (6)
Policy 1, threshold c_p/c_c	18.9 (4)	48.8 (5)	23.4 (6)	39.0 (4)	77.2 (20)
Cost-tuned STW, 1-SE	4.4 (0)	6.1 (0)	5.3 (0)	18.1 (1)	17.8 (2)
Age replacement	64.2 (1)	79.3 (1)	73.5 (2)	73.0 (0)	91.7 (1)

replacements per failure, and CNA-DA's M rose to 50–159 %, close to that of policy 1 (27–156 %), against 4–61 % for the one-standard-error rule. The comparison is in our implementation of their policy, with their probability model replaced by an empirical one built from our regressor, and on our splits; the values they report come from other models on a single 20-engine split and are not comparable.

7.6. Transfer: a calibrated level does not survive a new batch; online calibration restores the failure rate

With a whole battery batch held out, calibration and test units are no longer exchangeable, and the guarantee failed as the protocol predicted (H5a). Calibrated on the other two batches, CNA at $\alpha = 0.10$ failed on 0 % of batch 1 but on 74 % of batch 2 and 61 %

of batch 3 (Table 6, Fig. 7). The shortest-lived batch was the worst, the direction expected when a regressor trained mostly on longer-lived cells overestimates remaining life. The same loss of validity appeared on N-CMAPSS with a flight-data subset held out: 29–40 % failures on three of nine subsets and 0–2 % on the other six. Conservative uncertified rules did better under this shift: the one-standard-error cost-tuned alarm failed on 0–2 % of the held-out battery cells and the capacity-trend policies on none. Their extra margin, which cost them little within batches, protected them across batches; a calibrated alarm has, by design, no margin beyond what its calibration units require.

Two remedies were examined. *Pilot recalibration* (H5b) used 10 randomly chosen cells of the held-out batch as calibration units and tested on the rest: the mean failure rate over 300 draws was 7.5–10.1 % at $\alpha = 0.10$ across batches, lead times and repetitions, around the exact value $1/11 = 9.1$ % of Proposition 1, with 48–60 cycles of notice; five pilot cells cannot certify $\alpha = 0.10$ at all. Its drawback is that the pilot cells must reveal their critical levels: a cell replaced under a conservative interim level can confirm that level but never relax it (Theorem 2). *Online calibration* (H10) needs no pilot failures. Starting from the level of the other batches and updating it by (3) after each cell's outcome ($\eta = 0.1$ of the signal cap, one cell at a time, 200 random orders), the failure rate over the whole

Table 6

Failure before replacement with a whole group held out (battery: lead time 30 cycles; N-CMAPSS: 10 cycles), mean of three repetitions. Static: CNA at $\alpha = 0.10$ calibrated on the other groups. Pilot: CNA calibrated on m randomly chosen units of the held-out group and tested on the rest (mean over 300 draws); with $m = 5$, $\alpha = 0.10$ is not attainable and $\alpha = 0.20$ is shown. Online: level updated after each unit's outcome ($\eta = 0.1$ cap, $g = 1$; mean over 200 orders); second half of the sequence in brackets.

Held-out group (units)	CNA 0.10, static	Online (2nd half)	Pilot $m = 10$, $\alpha = 0.10$	Pilot $m = 5$, $\alpha = 0.20$	CNA-DA, static	STW window	STW 1-SE	Kamariotis P1 opt.	Trend 1-SE
Battery batch 1 (36)	0.00	0.076 (0.104)	0.095	0.167	0.00	0.00	0.00	0.00	0.00
Battery batch 2 (43)	0.74	0.127 (0.100)	0.094	0.165	0.45	0.35	0.00	0.12	0.00
Battery batch 3 (44)	0.61	0.135 (0.103)	0.084	0.167	0.36	0.06	0.02	0.17	0.00
N-CMAPSS, 6 subsets (9–15 each)	0.00–0.02	0.000–0.055	–	0.165–0.177	0.00	0.00–0.02	0.00	0.00	–
N-CMAPSS subset 4 (10)	0.33	0.206 (0.159)	–	0.165	0.10	0.07	0.10	0.07	–
N-CMAPSS subset 8 (15)	0.29	0.178 (0.146)	–	0.169	0.18	0.11	0.18	0.11	–
N-CMAPSS subset 9 (10)	0.40	0.192 (0.135)	–	0.169	0.07	0.10	0.13	0.03	–

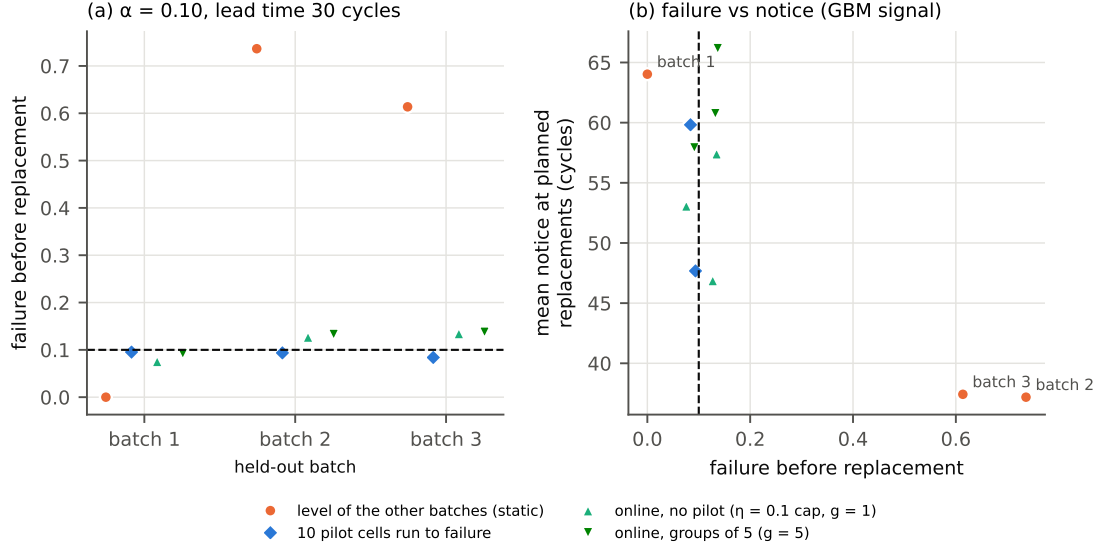


Fig. 7: Battery cells with a whole batch held out ($\alpha = 0.10$, lead time 30 cycles). (a) Failure before replacement with the level calibrated on the other batches (static), recalibrated on 10 pilot cells of the held-out batch, and updated online after each cell's outcome (one cell or five cells at a time). (b) Failure against mean notice at planned replacements.

sequence of held-out cells was 7.6, 12.7 and 13.5 % on batches 1–3 (mean of three repetitions; the bound of Theorem 4 held in every one of the 316,800 simulated sequences) and 10.0–10.4 % over the second half of each sequence, with 47–57 cycles of notice at planned replacements. On batch 1, where the static level had been conservative, the online level relaxed and gave 11 cycles less notice. Updating after groups of five cells, as for units operated concurrently, gave 9.1–13.7 %; a smaller step ($\eta = 0.05$ cap) adapted more slowly (4.8–16.4 %), a larger one (0.2) faster but with more variable notice. On the N-CMAPSS subsets, whose sequences are only 9–15 engines long, the online level lowered the failure rate on the three shifted subsets from 29–40 % to 18–21 %, but such short sequences leave too little room for adaptation; Theorem 4 then gives only a weak bound (0.40–0.75).

7.7. What the window score measured

Proposition 2 says that the window scores of the estimate and of a constant can differ only through units

whose notice falls near an edge of the constant's acceptance band. On every dataset the reported estimate was pinned below the level as part (iii) states (100 % of alarms within $[H - k\delta^+, H]$) and varied much less than the notice: its standard deviation was 2.3–3.2 cycles on the engines against 4.9–18.0 for the notice (battery 6.3 against 12.9). Discordant units, correct under one report and not the other, were 6.9–14.0 % on PHM08 and C-MAPSS, 11.4 % on the battery cells and 30.3 % on N-CMAPSS (Table 7).

The equivalence test of the protocol was not supported. With a margin of ± 5 points, estimate and constant were equivalent only on FD002; on FD001, FD003 and N-CMAPSS the intervals were too wide to decide; on PHM08 and FD004 the constant was significantly *better* than the model's estimate (–6.4 and –6.0 points), and on the battery cells the estimate was significantly *better* (+9.8 points), the one dataset where the estimate's spread at the alarm was a sizeable fraction of the window width (6.3 of 45 cycles). This corrects the earlier summary that the estimate changed the score by at most 3.6 points [25]. The plug-in bound

Table 7

Window score of the window-tuned alarm's estimate and of one constant at the same alarm times ($H = 20$ cycles; battery 60), repetition 0. Difference: estimate – constant with Tango 90 % interval; equivalence margin ± 5 points. SD at the alarm in cycles. Bound: Proposition 2(ii) estimated on calibration units; Disc.: realised share of discordant units.

Dataset	n	Estimate (%)	Constant (%)	Discordant (est./const.)	Difference (90 % CI)	TOST p	SD estimate / notice	Bound / Disc. (points)
PHM08	218	87.6	94.0	4 / 18	−6.4 (−10.3, −3.1)	0.756	2.6 / 4.9	15.9 / 10.1
FD001	100	86.0	85.0	6 / 5	+1.0 (−4.9, +7.0)	0.126	3.0 / 5.3	19.1 / 11.0
FD002	260	90.4	91.9	7 / 11	−1.5 (−4.5, +1.2)	0.028	2.4 / 5.2	13.6 / 6.9
FD003	100	84.0	86.0	6 / 8	−2.0 (−8.6, +4.5)	0.216	2.9 / 5.3	13.6 / 14.0
FD004	249	79.1	85.1	6 / 21	−6.0 (−9.7, −2.7)	0.694	2.3 / 18.0	15.2 / 10.8
N-CMAPSS	99	52.5	56.6	13 / 17	−4.0 (−13.3, +5.2)	0.431	3.2 / 11.0	28.1 / 30.3
Battery	123	95.1	85.4	13 / 1	+9.8 (+5.4, +15.3)	0.964	6.3 / 12.9	8.9 / 11.4

of part (ii), computed on calibration units, exceeded the realised score difference on 6 of 7 datasets and the realised discordance on 4 of 7; it uses estimated probabilities and is not a guarantee. For this article the conclusion is the one that motivates the method: a window score bounds neither the notice nor the failure probability, so neither should be inferred from it.

8. Discussion

8.1. What the results establish

Calibrating the alarm rather than the interval did what Theorem 1 says it does. On seven datasets, with between 32 and 83 calibration units, the failure-before-replacement rate followed its nominal level, while a per-cycle lower bound at the same nominal level, on the same signal and trigger, failed on up to 70 % of units; the contrast held with four learners and the guarantee with 27 alarm settings. The difference is not a matter of tuning: per-cycle coverage averages over cycles chosen independently of the data, whereas an alarm is read at the first cycle at which the signal crosses a level, the cycle at which an optimistic error is most likely. Restricting the per-cycle residuals to the near-failure region repaired most of the gap for GBM on six datasets, but how close it came depended on that choice of rows and on the learner: on N-CMAPSS it exceeded its level by half, and with ridge regression it failed on 17–33 % of units. Calibrating the critical level removes the choice, because the quantity calibrated is the event the operator cares about. The practical message for the growing use of conformal RUL intervals in maintenance [4–6] is that interval coverage should not be read as alarm reliability.

8.2. A deployment recipe

Table 8 turns the results into a procedure. Its steps are the ones that, in this study, changed whether the stated reliability was achieved; none requires a model of the degradation or of the RUL error.

8.3. What the guarantee costs

The guarantee is a certificate; whether it also saves money depends on how many calibration units it rests on. Proposition 1(ii) gives the mechanism: a level

Table 8

Deployment recipe for a notice-guaranteed end-of-life alarm.

Step and reason	
1	State the lead time ℓ and the tolerated probability α of failure before the planned replacement; check $n \geq 1/\alpha - 1$ calibration units are available (Proposition 1); if the requirement must hold for the calibrated alarm itself with confidence $1 - \delta$, choose the order statistic by Corollary 1.
2	Fix the signal and trigger (smoothing, debounce, cap) before looking at the calibration units, or select them on units not used for calibration.
3	Collect calibration units from the population in which the alarm will run. Units replaced late in life, for example by an earlier alarm, can serve as censored calibration units, their critical level computed as if they had failed immediately after they were last seen alive (Theorem 2); units censored early in life carry almost no information.
4	Compute critical levels (Lemma 2) and set $\hat{H}$ to their conformal quantile; with a cost ratio, choose among admissible order statistics (Corollary 2).
5	When the population changes (new batch, campaign or operating regime), do not carry a calibrated level across: update it online from the new units' outcomes (Theorem 4), or recalibrate on units of the new population that were run close to failure.
6	Report the failure rate with its interval, the notice given, and the number of calibration units the claim rests on; do not report interval coverage or a window score as alarm reliability.

chosen among n calibration critical levels fails with probability at least $1/(n + 1)$, and with expensive failures that floor dominates the cost. With the split design and 32 calibration units the certified alarm was clearly dearer than the best uncertified rules; cross-fitting, which calibrates on all training units and trains the signal on four fifths of them, removed most of the premium (Table 4). With equal information, cross-fitted calibration made the certificate almost free: the certified alarm was never significantly dearer than the best uncertified rules built on the same models, and cheaper than the one-standard-error rule on two datasets (Section 7.4). Rules that choose levels beyond the most demanding calibration unit can be cheaper on a given test set, and the earlier study showed how such rules can fail when plain empirical cost minimisation finds a setting at the edge of the feasible region [25]; what they cannot provide is a bound for the next unit. Two sizing rules follow. To certify a failure probability

α one needs at least $1/\alpha - 1$ run-to-failure (or late-censored) calibration units from the population in question; to keep the certified excess cost $(\rho - 1)/(n + 1)$ below a fraction ε of a planned replacement one needs $n + 1 \geq (\rho - 1)/\varepsilon$. Where such data exist, as for fleets of identical components or cells from one production line, the certificate is cheap; where they do not, a failure rate of an uncertified alarm measured on a few dozen units is weak evidence of its reliability.

8.4. Transfer and censoring in the field

The battery results show the limit of every exchangeability-based guarantee: calibrated on two batches, the alarm failed on most cells of a third, shorter-lived batch, and similar failures occurred on three of nine N-CMAPSS subsets. This is not a defect of the conformal step, which did what it promised for the population it was calibrated on, but it is a warning against reading a validated failure rate as a property of the alarm rather than of the alarm and a population. Three responses were examined. Conservative uncertified rules carried enough margin to survive this shift, at the price of notice that is wasted when the population does not change. Recalibration on ten pilot cells of the new batch restored the per-unit guarantee, but those cells must be run close to failure, because a cell replaced under a conservative level can confirm that level but never relax it. Online calibration (Theorem 4) needs neither exchangeability nor pilot failures: it uses only whether each new unit failed before its replacement, which is observed in any fleet, and it brought the failure rate of the shifted batches back to about 10% after the first half of each sequence. Its guarantee is of a different kind, a bound on the realised long-run rate rather than a probability for the next unit, and short sequences, early failures in a strongly shifted population and delayed feedback limit it. Weighted or non-exchangeable conformal methods [21, 22] could use covariates of the new batch, such as its charging protocol, to reweight calibration units; this needs a model of the shift and was not attempted.

8.5. Limitations

- **Data.** Five of the seven datasets come from one simulator family and the battery cells are laboratory data. Censoring was imposed on complete run-to-failure records by three mechanisms; no field fleet with genuinely censored lives was analysed, and none was available to us, nor was a second laboratory dataset of the same cell type that would test transfer across laboratories. Bearings were excluded because 17 units cannot support the guarantee.
- **Exchangeability.** The per-unit guarantees require exchangeable calibration and new units; they failed across battery batches and N-CMAPSS subsets and may fail under slower drifts within one population. The online update replaces them by a bound on

the realised long-run failure rate that holds for any sequence but says nothing about an individual unit, is weak for short sequences and assumes that outcomes are fed back promptly. Theorem 1 is marginal over calibration sets; the training-conditional form (Corollary 1) needs independent units, and making it hold with high confidence costs notice.

- **Scope.** The theory covers threshold triggers with debounce on a causal signal and a fixed lead time. The settings of CNA (cap, smoothing, debounce) were fixed a priori, not optimised; 27 alternative settings gave pooled failure rates of 8.7–11.1% and almost the same notice. The cross-fitted guarantee (Theorem 3) is weaker than the split guarantee.
- **Cost.** The renewal–reward model assumes independent units, replacement as good as new and illustrative cost ratios. Section 7.5 uses the decision setting of Kamariotis et al. but our own implementation of their policy and of the probability model behind it.
- **Pre-specification.** The protocol and its four amendments were hashed before their runs but not registered, and each amendment was written after earlier results were known. The equivalence and cost hypotheses of the main protocol and the cost hypothesis of the first amendment (H7b) failed and are reported as such; the prominence of Proposition 1(ii) as the explanation of the cost results was decided after those results were seen, and the check of Corollary 1 was added after all runs.

9. Conclusion

An end-of-life alarm is useful when it leaves enough notice for the replacement it schedules. This article calibrated that property directly. For a threshold trigger the notice is monotone in the warning level, every calibration unit has a single critical level, and the conformal quantile of those levels gives a warning level whose probability of failure before the planned replacement is at most α for a new exchangeable unit. The guarantee extends to calibration data censored in any way, at a price that depends on how early units were censored, to cross-fitted calibration on all training units, and, in a long-run sense, to populations that change. A decision-aware choice keeps it, and the number of calibration units sets a floor, $1/(n + 1)$, on the failure probability any such choice can certify.

On seven run-to-failure datasets and with four different learners the guarantee held where it should and failed where it should not be expected to hold. Per-cycle conformal lower bounds on the same signal failed on up to 70% of units at nominal 10%, so interval coverage should not be read as alarm reliability. Calibration histories produced by an earlier alarm were as good as complete data. With cross-fitting, certification cost about as much as the best uncertified tuning rules

built on the same models. A calibrated level did not survive a change of battery batch; ten pilot cells of the new batch restored it, and so did an online update from the batch's own outcomes, without any pilot failure. The practical recommendation is to state the failure probability an alarm is meant to achieve, to calibrate it on units of the population in which it will run, using replaced units where they were replaced late, and to report how many units the claim rests on. Open problems are a per-unit guarantee under a changing population without pilot failures, online updating with long feedback delays, and validation on field fleets with genuinely censored lives.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to search the literature, write and run the analysis code, draft the theory and proofs, and draft and revise the manuscript text. The protocols, all experiments and all reported numbers were produced by the code in the accompanying repository and checked against its output files. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data and code availability

PHM08, C-MAPSS and N-CMAPSS are available from the NASA Prognostics Center of Excellence data repository, and the battery data of Severson et al. [28] from <https://data.matr.io>. Code, hashed protocols, derived data summaries, per-unit results and the scripts that produce every table and figure are in a repository whose link is withheld for anonymised review.

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